

Noushin Omidvar

Applied Machine Learning Scientist | Email: noush.omidvar@gmail.com | LinkedIn: [noushin-omidvar](https://www.linkedin.com/in/noushin-omidvar)

SUMMARY

Applied ML scientist who slashes preprocessing from days to hours and speeds up screening 40% - proven track record turning complex data into deployed business value.

PROFESSIONAL SKILLS

Languages & Tools: Python, SQL, Git, Jupyter

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn

ML Frameworks & Libraries: Scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch

Experimentation & Analysis: Weights & Biases, Optuna, SHAP, LIME

Cloud Platforms: AWS (EC2, S3), Google Cloud Platform

WORK EXPERIENCE

Machine Learning Scientist,

Aionics Inc., Palo Alto | 2024–Present

Predictive Modeling for Electrolyte and Molecular Formulations Properties

- Developed and deployed ML models to predict electrolyte and molecular formulation properties, **boosting candidate screening speed by 40%**.
- Automated high-throughput data curation and feature engineering pipelines, **reducing preprocessing time from days to hours**.
- Applied explainable AI methods (SHAP, diagnostics) to improve model interpretability, **increasing adoption across cross-functional teams**.
- Standardized experiment tracking with Weights & Biases, ensuring **reproducibility and workflow alignment** across engineering and science teams.

Multi-Objective Optimization for Experimental Design

- Designed and implemented optimization workflows to balance conductivity, viscosity, and safety constraints in electrolyte formulations.
- Prioritized experimental candidates via Pareto-front selection, **cutting R&D iteration cycles by 60%**.
- Collaborated with chemists and engineers to integrate optimization outputs into lab workflows, **reducing trial-and-error costs**.

ACADEMIC RESEARCH EXPERIENCE

Ph.D. Researcher, Machine Learning for Scientific Discovery,

Virginia Tech, Department of Chemical Engineering | 2016–2022

Interpretable AI for Catalysis and Materials Discovery

- Built Python-based ML pipelines integrating DFT-calculated properties with custom surface descriptors, **enabling large-scale screening of novel materials**.
- Developed interpretable ML models to analyze chemical bonding and catalytic activity, **advancing design principles for energy conversion catalysts**.
- Applied statistical modeling to uncover physical-chemical drivers of performance, **guiding experimental validation of high-performing catalysts**.
- Published **6+ peer-reviewed articles** in top journals (*Nature Communications*, *Trends in Chemistry*, *J. Phys. Chem. Letters*), establishing expertise in interpretable AI.
- Mentored graduate researchers on ML modeling and data analysis, **increasing team productivity and knowledge transfer**.

SELECTED Publications

- Omidvar, N. et al. "Explainable AI for Optimizing Oxygen Reduction on Pt Catalysts." *Electrochemical Science Advances*, 2024
- Huang, Y., Omidvar, N. et al. "Reactivity Origin of Oxygen Reduction at High-Entropy Alloys." *J. Phys. Chem. C*, 2024
- Pillai, H.S., Omidvar, N. et al. "Interpretable Design of Electrocatalysts with Graph Neural Networks." *Nature Communications*, 2023
- Omidvar, N. et al. "Interpretable Machine Learning of Chemical Bonding at Solid Surfaces." *J. Phys. Chem. Letters*, 2021
- Omidvar, N. & Xin, H. "Algorithm-derived Feature Representations for Explainable AI in Catalysis." *Trends in Chemistry*, 2021
- Li, Z., Omidvar, N. et al. "Machine Learning Energy Gaps of Porphyrins." *J. Phys. Chem. A*, 2018

Full list on [Google Scholar](#)

EDUCATION

Ph.D. in Chemical Engineering - Virginia Tech, USA | 2022

M.Sc. in Chemical Engineering - Tarbiat Modares University, Iran | 2015

B.Sc. in Chemical Engineering - Sharif University of Technology, Iran | 2012